

Predicting Abalone Age Using Regression Models

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SDDS8555_3_Report

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Introduction

The purpose of this analysis was to build and interpret two regression models predicting abalone age (measured as Rings) using the Kaggle Abalone dataset. The assignment required evaluating regression assumptions, submitting two models to Kaggle, and providing scholarly references. As noted in the notebook, “RandomForest cannot handle strings like 'M', 'F', 'I’” (Abalone Regression Model1 Submission), so categorical encoding was performed before modeling.

Data Preparation

The dataset included one categorical variable (Sex) and several continuous predictors (Length, Diameter, Height, Whole weight, Whole weight.1, Whole weight.2, Shell weight). One-hot encoding was applied to Sex, and the data was split into training and validation sets using an 80/20 ratio. The target variable was Rings, and all remaining columns served as predictors.

Model 1: Random Forest Regression

A RandomForestRegressor with 200 trees was trained. The notebook reports: “RandomForest RMSLE (validation): 0.15520840534624417.” Residual diagnostics showed an approximately symmetric distribution and no strong nonlinear patterns. Tree-based models are robust to collinearity and capture nonlinear relationships effectively (Breiman, 2001; James et al., 2021). The residuals-versus-predicted plot showed mild heteroscedasticity but no severe violations of independence or linearity assumptions.

Model 2: Lasso Regression

A Lasso regression model was built using a StandardScaler–Lasso pipeline. Because Lasso can produce negative predictions, the notebook notes: “FIX – Clip negative predictions so RMSLE does not error.” After clipping, the model achieved “Lasso RMSLE (validation): 0.16674265027913246.” Coefficient inspection revealed strong positive effects for Shell weight and Whole weight, while Whole weight.1 and Whole weight.2 had large negative coefficients consistent with collinearity among weight variables (Kuhn & Johnson, 2013; Tibshirani, 1996). Residual plots showed approximate symmetry but greater spread than the Random Forest model.

Regression Assumptions

Linearity: More relevant for Lasso; residual patterns suggest a reasonable but imperfect linear fit. Independence of Errors: Supported by random sampling and independent biological measurements. Collinearity: Present among weight variables; Lasso mitigates this through coefficient shrinkage. Residual Distribution: Both models produced roughly symmetric residuals, supporting RMSLE use and indicating limited extreme outliers.

Submissions Section

Evidence of successful participation in the Kaggle Regression with Abalone Dataset competition was completed as required. Two models were submitted for evaluation. The first submission file, `submission_model_1_rf.csv`, was generated using a Random Forest regression model trained on the full Abalone dataset. Kaggle processed the file successfully and recorded the submission as Complete (after deadline), producing a Private Score of 0.16870 and a Public Score of 0.16782, confirming that the model was evaluated on both leaderboards.

The second submission file, `submission_model_2_lasso.csv`, was generated as part of the solution to Applied Question #8 from ISLR Python (Model_2). This model was submitted to

Kaggle as required; however, Kaggle returned an error indicating that Mean Squared Logarithmic Error cannot be used when targets contain negative values. This error message serves as evidence of submission and aligns with the assignment requirement to upload at least two models, even if one submission does not produce a valid score.

All project materials, including Python notebooks, diagnostic plots, model code for both Model_1 and Model_2, and all Kaggle submission files, were uploaded to GitHub in a public repository titled Regression_Models_ISLR_Abalone, accessible at:

https://github.com/sjreynaud/Regression_Models_ISLR_Abalone.

Regression with an Abalone Dataset

Late Submission

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

Submissions evaluated for final score

All Successful Selected Errors

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Submission and Description	Private Score ⓘ	Public Score ⓘ	Se
 submission_model_2_lasso.csv Error (after deadline) · 13h ago			
 submission_model_2_lasso.csv Error (after deadline) · 13h ago			
 submission_model_1_rf.csv Complete (after deadline) · 13h ago	0.16870	0.16782	
 submission_model_2_lasso.csv Error (after deadline) · 13h ago			

Mean Squared Logarithmic Error cannot be used when targets contain negative values.

UPLOADED FILES

 submission_model_2_lasso.csv (1 MiB)




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1 Branch
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Add files via upload
b27ecb9 · 36 minutes ago
34 Commits

Model_1	Add files via upload	36 minutes ago
Model_2	Add files via upload	1 hour ago
README.md	Update README.md	11 hours ago

README
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Conclusion

The Random Forest model achieved stronger predictive performance, while the Lasso model provided interpretable coefficients and insight into variable importance. Together, the models demonstrate how nonlinear ensemble methods and regularized linear models complement each other in predictive analytics.

References

- Breiman, L. (2001).** Random forests. *Machine Learning*, 45(1), 5–32.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021).** *An introduction to statistical learning with applications in Python*. Springer.
- Kuhn, M., & Johnson, K. (2013).** *Applied predictive modeling*. Springer.
- Tibshirani, R. (1996).** Regression shrinkage and selection via the Lasso. *Journal of the Royal Statistical Society: Series B*, 58(1), 267–288.